Model Optimization & Deployment Documentation

Gurgaon Real Estate Property Valuation Model

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1. Executive Summary

This document provides a complete overview of the **hyperparameter tuning, validation, optimization, and production serialization workflow** developed for the Gurgaon Real Estate Property Valuation Model.

The primary objective of this notebook was to build a high-performance regression system capable of accurately predicting property prices while maintaining strong generalization on unseen market data.

The optimization strategy focused on:

- Minimizing **Mean Absolute Error (MAE)**
- Maximizing **R^2 (Coefficient of Determination)**
- Reducing overfitting through regularization
- Building a production-ready inference pipeline

An automated hyperparameter optimization tournament using **Optuna** was conducted across multiple model configurations. The final champion model was an **XGBoost Regressor**, selected due to its superior ability to capture complex nonlinear relationships between property attributes and market prices.

The final pipeline successfully learned complex pricing interactions, including:

- Premium differences between independent houses and apartments
- Sector-wise price variations
- Impact of property age and possession status
- Luxury category influence on valuation

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2. Feature Preprocessing Architecture

To prevent data leakage and ensure consistent production inference, all feature transformations are encapsulated inside a unified `ColumnTransformer` pipeline.

The preprocessing framework contains four major transformation layers.

2.1 Numerical Feature Scaling

Selected numerical variables:

```
[
    'bedRoom',
    'bathroom',
    'built_up_area',
    'servant room',
    'store room'
]
```

Transformation:

```
StandardScaler
```

Purpose:

- Normalize feature distributions
- Reduce scale imbalance
- Improve model optimization stability

2.2 Ordinal Feature Encoding

Ordinal categorical variables are transformed using:

```
OrdinalEncoder(
    handle_unknown='use_encoded_value',
    unknown_value=-1
)
```

Advantages:

- Supports unseen production categories
- Prevents inference failure
- Maintains ordered category representation

2.3 Nominal Feature Encoding

The `agePossession` feature is transformed using:

```
OneHotEncoder(
    sparse_output=False,
    handle_unknown='ignore'
)
```

Purpose:

- Convert categorical states into binary indicators
- Preserve structural property characteristics
- Avoid prediction errors from unseen values

2.4 Spatial Target Encoding

High-cardinality geographic information:

```
sector
```

is processed using localized target encoding.

Benefits:

- Captures regional pricing patterns
- Reduces dimensional explosion
- Provides location-based valuation signals

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3. Target Transformation Strategy

Real estate prices naturally follow a highly right-skewed distribution because luxury properties create extreme price variations.

To stabilize training, the target variable was transformed using:

$$f(x) = \ln(1 + x)$$

where:

- `x` = actual property price in Crores

The transformation was implemented using:

```
TransformedTargetRegressor
```

During prediction, values are automatically converted back:

$$f^{-1}(x) = e^x - 1$$

This ensures:

- Training happens in a stable logarithmic space
- Evaluation remains in real price units
- Production predictions are directly usable

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4. Champion Model Architecture

After multiple optimization generations, **XGBoost Regression** achieved the best validation performance.

The final architecture uses:

- Deep learning of nonlinear interactions
- Tree regularization
- Feature subsampling
- Row subsampling
- Controlled learning rate

Final Hyperparameter Configuration

```
champion_hyperparameters = {
    'n_estimators': 500,
    'learning_rate': 0.020549749861957282,
    'max_depth': 11,
    'subsample': 0.8266692560047036,
    'colsample_bytree': 0.7629300527334719,
    'reg_alpha': 0.017773044792202236,
    'reg_lambda': 0.9447599108174097,
    'random_state': 42,
    'n_jobs': -1
}
```

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5. Model Performance Evaluation

The final model was evaluated using a completely unseen holdout validation dataset.

Dataset split:

Dataset	Samples
Training Data	2843
Testing Data	711

Final Holdout Metrics

Metric	Score	Interpretation
MAE	0.4192 Crores	Average prediction error is approximately 41.9 Lakhs
R^2 Score	0.8320	Model explains 83.2% of market price variance

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6. Generalization Verification

The final production pipeline demonstrated strong stability.

Validation observations:

- Holdout MAE remained below the internal Optuna CV benchmark (~0.47 Crores)
- Performance gap between training and validation remained controlled
- Regularization successfully reduced variance
- Model avoided significant overfitting

The final model is considered production-ready.

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7. Production Inference Pipeline

All preprocessing and transformations are stored inside the trained pipeline object.

Therefore, production systems require:

- No manual feature transformation
- No duplicate preprocessing logic

- No separate encoding scripts

The backend only needs to provide raw property information.

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8. Model Serialization

Save the trained production pipeline:

```
import joblib

joblib.dump(
    final_production_pipeline,
    'gurgaon_property_valuation_champion.joblib'
)
```

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9. Production Loading & Prediction

```
import joblib
import pandas as pd


deployed_pipeline = joblib.load(
    'gurgaon_property_valuation_champion.joblib'
)


live_query = pd.DataFrame([{
    'property_type': 'house',
    'sector': 'sector 102',
    'bedRoom': 4,
    'bathroom': 3,
    'balcony': '3+',
    'agePossession': 'New Property',
    'built_up_area': 2750,
    'servant room': 0,
    'store room': 0,
    'furnishing_type': 'unfurnished',
    'luxury_category': 'Standard',
    'floor_category': 'Low Floor'
}])


```

```python
valuation = deployed_pipeline.predict(
    live_query
)[0]


print(
    f"Verified Real-Time Valuation: {valuation:.4f} Crores"
)
```

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10. Deployment Readiness Checklist

Component	Status
Feature preprocessing pipeline	✅ Completed
Leakage prevention	✅ Implemented
Hyperparameter optimization	✅ Completed using Optuna
Champion model selection	✅ Completed
Holdout validation	✅ Passed
Model serialization	✅ Completed
Production inference compatibility	✅ Verified

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11. Final Conclusion

The Gurgaon Real Estate Property Valuation Model represents a complete machine learning lifecycle:

1. Data preprocessing
2. Feature engineering
3. Target transformation
4. Automated hyperparameter optimization
5. Model validation
6. Pipeline serialization
7. Production inference

The final XGBoost-based architecture provides reliable property valuation with strong predictive accuracy and production-grade deployment capability.

Final Model Status: Production Ready 🚀